

Project Proposal

Project Title

IdeaForge AI: An AI-Powered Academic Project Intelligence Platform

1. Introduction

Selecting an appropriate academic project is one of the most challenging tasks for undergraduate students. Every academic year, students spend significant time searching for project ideas, only to discover that many of them have already been implemented by previous batches. At the same time, faculty members manually review project proposals to identify duplicated ideas, evaluate feasibility, and recommend improvements. This process is often time-consuming and lacks a centralized system for managing previous academic projects.

IdeaForge AI aims to solve these problems by providing an AI-powered platform that assists students in discovering project ideas, evaluating proposal originality, and receiving intelligent recommendations before submission. The platform also assists faculty by automating proposal analysis and providing valuable insights to support the review process.

Unlike traditional project repositories that simply store documents, IdeaForge AI functions as an intelligent academic assistant that leverages Artificial Intelligence and Retrieval-Augmented Generation (RAG) techniques to analyze previous projects and provide meaningful recommendations.

2. Problem Statement

Most universities maintain archives of completed projects; however, these archives are difficult to search and rarely assist students during project selection. As a result:

- Students unknowingly choose project ideas that have already been implemented.
- Students struggle to determine the feasibility and difficulty of their ideas.
- Faculty members manually compare new proposals with previous projects.
- Students receive limited guidance on suitable technologies and implementation approaches.
- There is no intelligent system that provides recommendations based on previous academic work.

These challenges reduce project originality and increase the workload for both students and faculty.

3. Objectives

The primary objective of IdeaForge AI is to develop an intelligent academic project management platform that improves the project selection and proposal evaluation process.

Specific objectives include:

- Develop a centralized repository for academic projects and proposals.
 - Enable AI-powered semantic search for previous projects.
 - Detect similarities between newly submitted proposals and archived projects.
 - Estimate project difficulty and implementation complexity.
 - Recommend suitable technologies based on project requirements.
 - Suggest potential faculty supervisors according to research interests.
 - Provide faculty with AI-assisted proposal evaluation and analytics dashboards.
 - Build a scalable platform that can expand as more projects are added.
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4. Proposed Solution

IdeaForge AI will provide separate portals for Students, Faculty Members, and Administrators.

Student Portal

Students will be able to:

- Search previous academic projects using natural language.
- Receive AI-generated project suggestions.
- Upload project proposals.
- View similarity scores with previous projects.
- Obtain novelty analysis.
- Receive technology recommendations.
- Track proposal status and faculty feedback.

Faculty Portal

Faculty members will be able to:

- Review submitted proposals.
- View AI-generated summaries.
- Analyze similarity reports.
- Approve, reject, or request modifications.
- Monitor student progress.
- View departmental project analytics.

Admin Portal

Administrators will manage:

- Student and faculty accounts.
 - Departments and academic sessions.
 - Project archives.
 - AI knowledge base.
 - Reports and system maintenance.
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5. AI Features

The intelligence of the platform will be achieved using Natural Language Processing (NLP) and Large Language Models (LLMs).

The proposed AI features include:

- Semantic project search
- Project similarity detection
- Novelty analysis
- Difficulty estimation
- Technology recommendation
- Supervisor recommendation
- AI-generated proposal summaries
- Retrieval-Augmented Generation (RAG) chatbot for answering project-related questions

Instead of generating random responses, the AI will retrieve relevant information from previous project reports and proposals before generating recommendations.

6. Methodology

The proposed development methodology consists of the following phases:

1. Requirement Analysis
2. System Design
3. Database Design
4. Frontend Development
5. Backend API Development
6. AI Module Development
7. Integration
8. Testing
9. Deployment
10. Evaluation

Agile software development practices will be followed throughout the project to allow continuous improvement.

7. Proposed Technology Stack

Frontend:

- React.js
- Bootstrap / Tailwind CSS

Backend:

- Django
- Django REST Framework

Database:

- PostgreSQL

Artificial Intelligence:

- Python
- LangChain or LlamaIndex
- Sentence Transformers
- FAISS / ChromaDB

- OpenAI API or Local LLM or Gemini API

Authentication:

- JWT Authentication

Version Control:

- Git & GitHub

Deployment:

- Docker (optional)
 - Render / Railway / VPS
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8. Expected Outcomes

The proposed system is expected to:

- Reduce duplication of academic projects.
 - Help students select more original and feasible project ideas.
 - Reduce proposal review time for faculty.
 - Improve communication between students and supervisors.
 - Build a continuously growing academic knowledge base.
 - Encourage innovation through AI-assisted recommendations.
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9. Scope of the Project

The initial version of IdeaForge AI will focus on project proposal management within a university department.

Future enhancements may include:

- Integration with university management systems.
 - Automatic research paper recommendations.
 - GitHub repository analysis.
 - AI-generated implementation roadmaps.
 - Industry mentor portal.
 - Mobile application.
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10. Conclusion

IdeaForge AI aims to modernize the academic project selection process by combining Artificial Intelligence with project management. The platform will provide students with intelligent guidance throughout project selection while reducing the workload of faculty members through automated proposal analysis. By maintaining a searchable repository of previous projects and using AI-powered recommendations, the system has the potential to improve project originality, enhance collaboration, and create a sustainable academic knowledge base for future students.